DEVK Meetup Köln, 03.07.2025

Me, myself and (Gen)AI, wie sich der Dev-Alltag verändert - Ein Erfahrungsbericht

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Wer bin ich?

- Sandra Parsick
- Freiberuflicher Softwareentwickler und Consultant im Java-Umfeld
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  - Open Source
  - Software Craftmanship
  - Automatisierung von Entwicklungsprozessen
- Trainings
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Support&Care for Apache Maven™



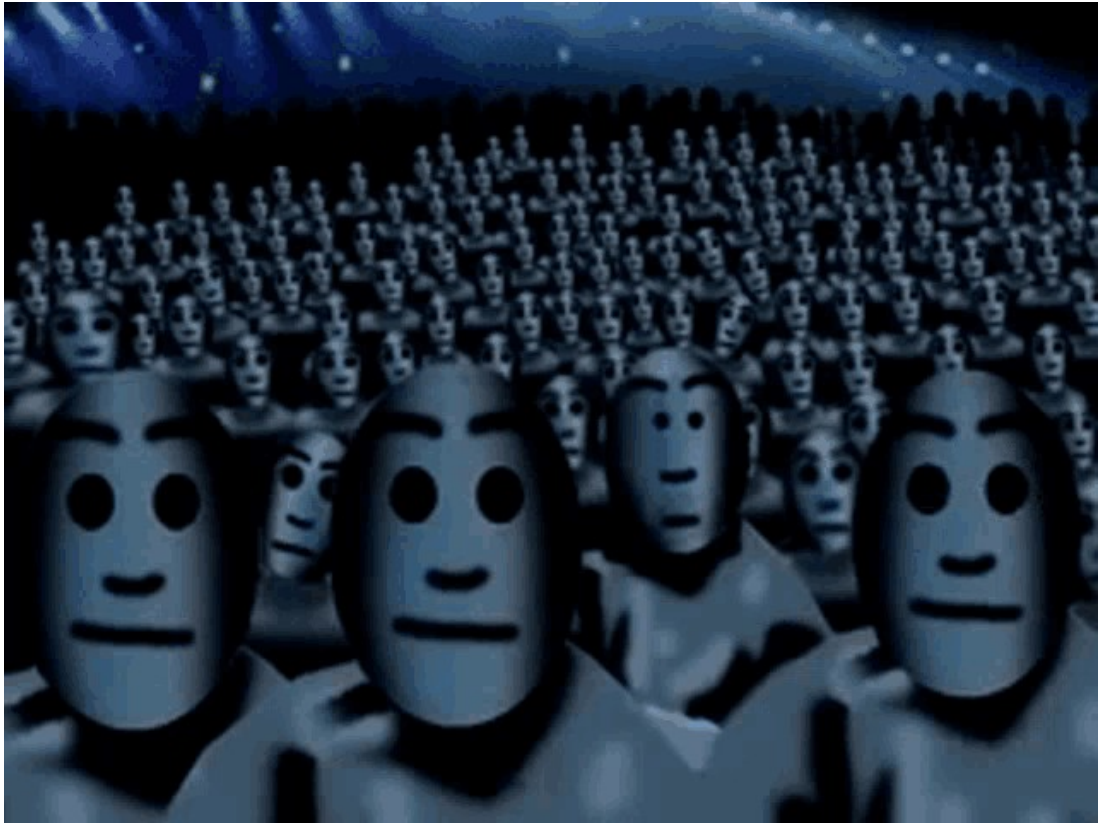
ACE Pro



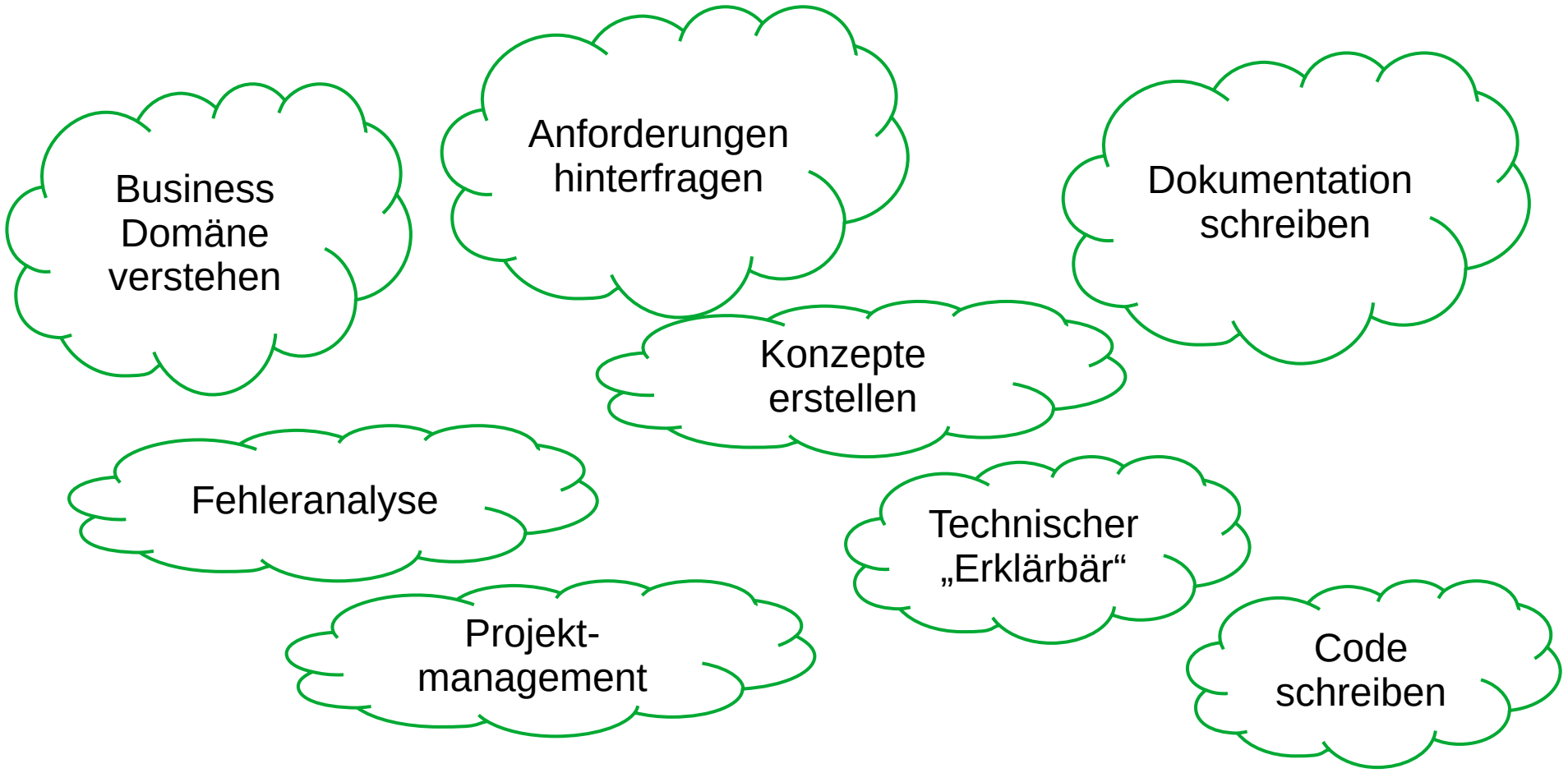


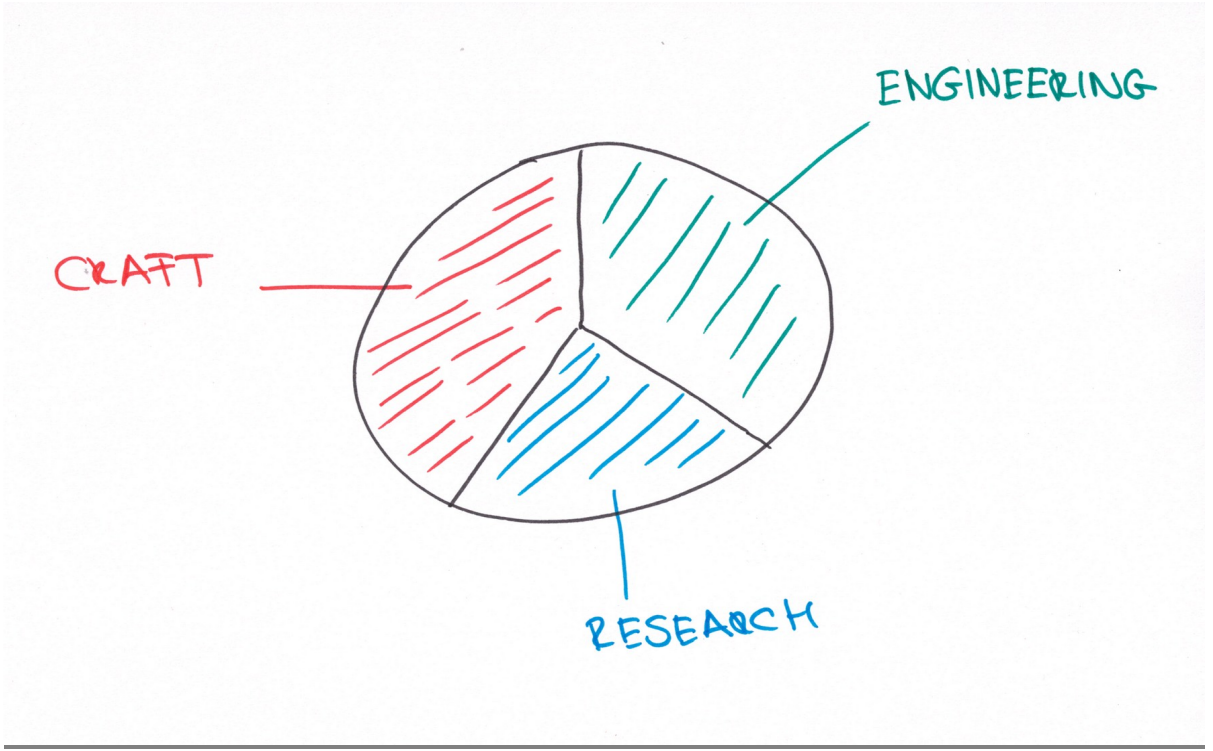
Testcontainers

Know your Audience



Mein Berufsalltag




ENGINEERING
CRAFT
RESEARCH

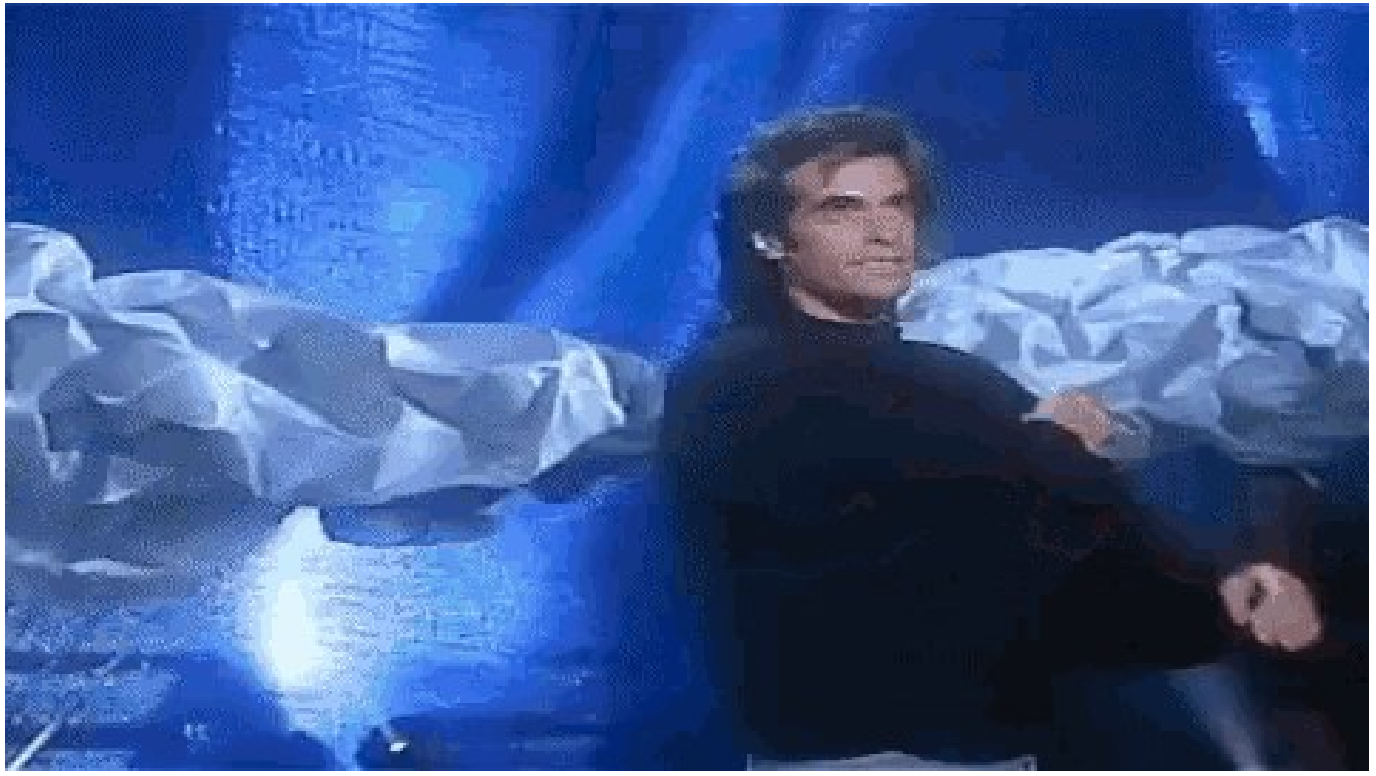
Agenda

- Gen AI, anhand LLM, kurz erklärt
- Wo mir LLM-basierte Tools NICHT helfen
- Wo mir LLM-basierte Tools helfen
- Auswirkungen im sozialen Umgang mit anderen Menschen

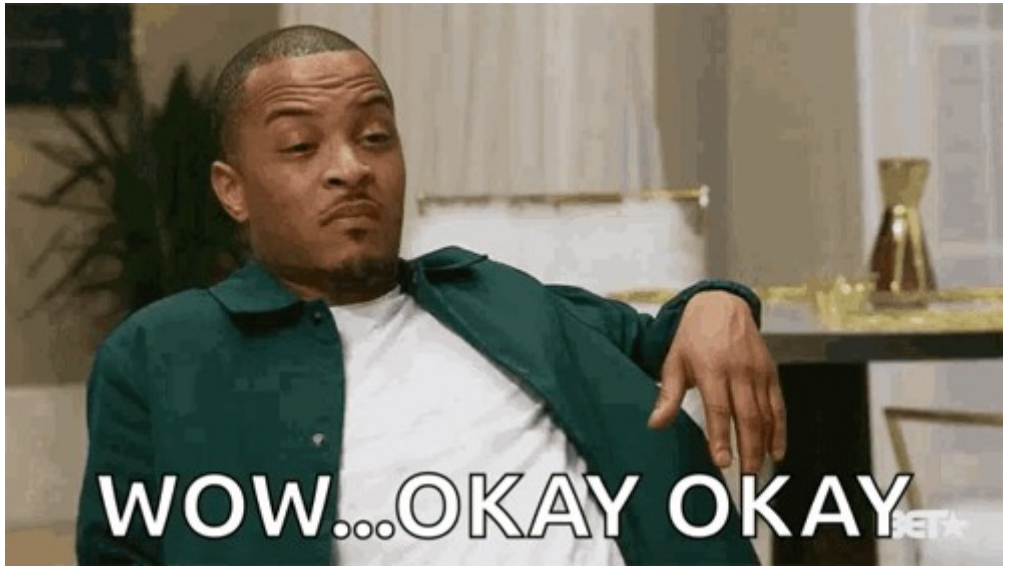
- Trigger Warnung
- Versuche ohne Ethik, Moral und Energieverbrauch auszukommen

Gen AI, anhand LLM, kurz erklärt

Warum ist es wichtig, die Technik zu verstehen?




WOW


WOW...OKAY OKAY


Breaking the Magician's Code . . .
MAGIC'S
BIGGEST SECRETS
FINALLY REVEALED!
INCLUDES
BONUS
FOOTAGE!
NEVER BEFORE
SEEN!
EXPOSED!

Vieles was wir nicht verstehen, hat etwas Magisches für uns



GenAI vs LLM

- GenAI = Generative KI
  - Oberbegriff für alle KI-Modelle, die Inhalt (Bild, Video, Audio, Text) erzeugen können
- LLMs = Large Language Models
  - Unterkategorie von GenAI, spezialisiert auf Textgenerierung

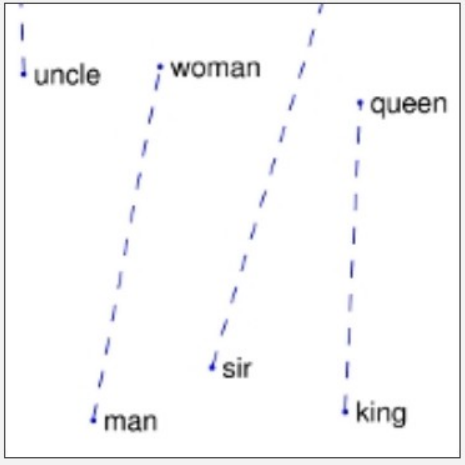
Funktionsweise LLMs

1) Tokenisierung (Tokenizer)
   Text in Token transformieren
2) Einbettung (Embedding)
   Token auf Vektoren abzubilden
3) Vorhersage
   Berechnung der Wahrscheinlichkeit des Tokens
4) Dekodierung
   Strategien zur Auswahl der Ausgaben

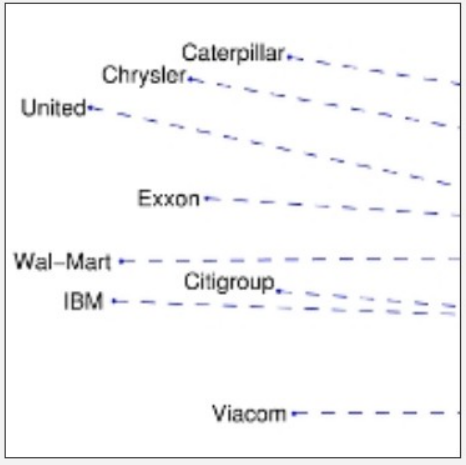
1) Tokenisierung



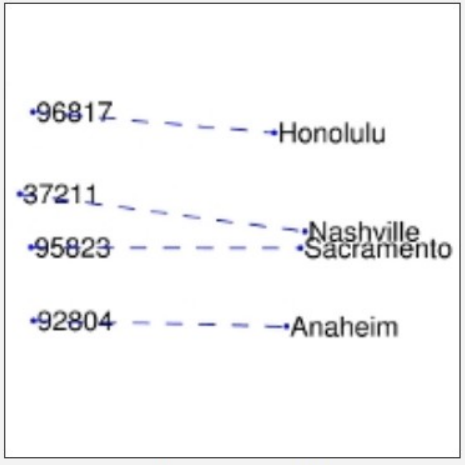
2) Token auf Vektoren abbilden



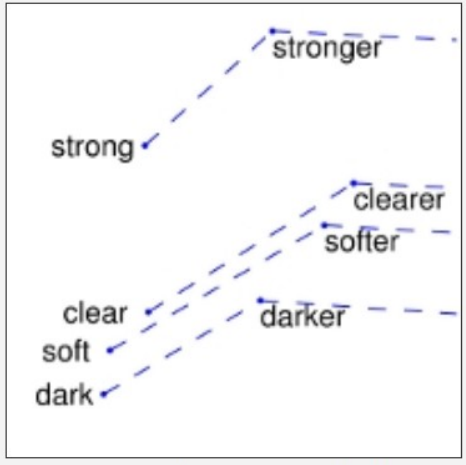
man - woman



company - ceo



city - zip code

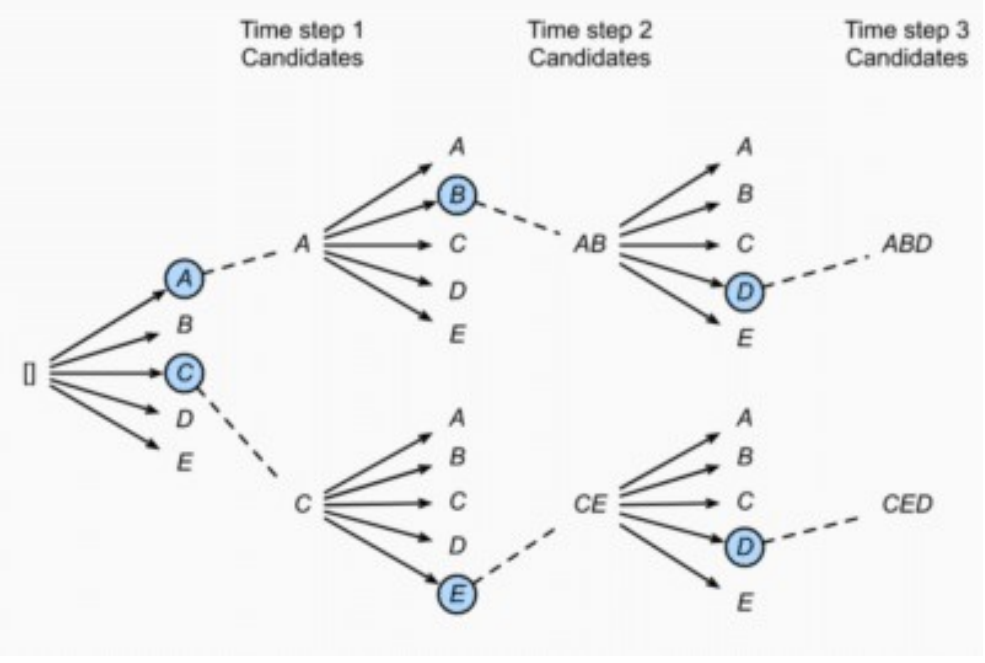


comparative - superlative

3) Vorhersage

- Kern des LLMs
- Verschiedene Architekturen
- Grundidee: Berechne die Wahrscheinlichkeit des nächsten Tokens

4) Dekodierung



Eine verständlichere Erklärung liefert die „Sendung mit der Maus“ zum Thema „Künstliche Intelligenz“

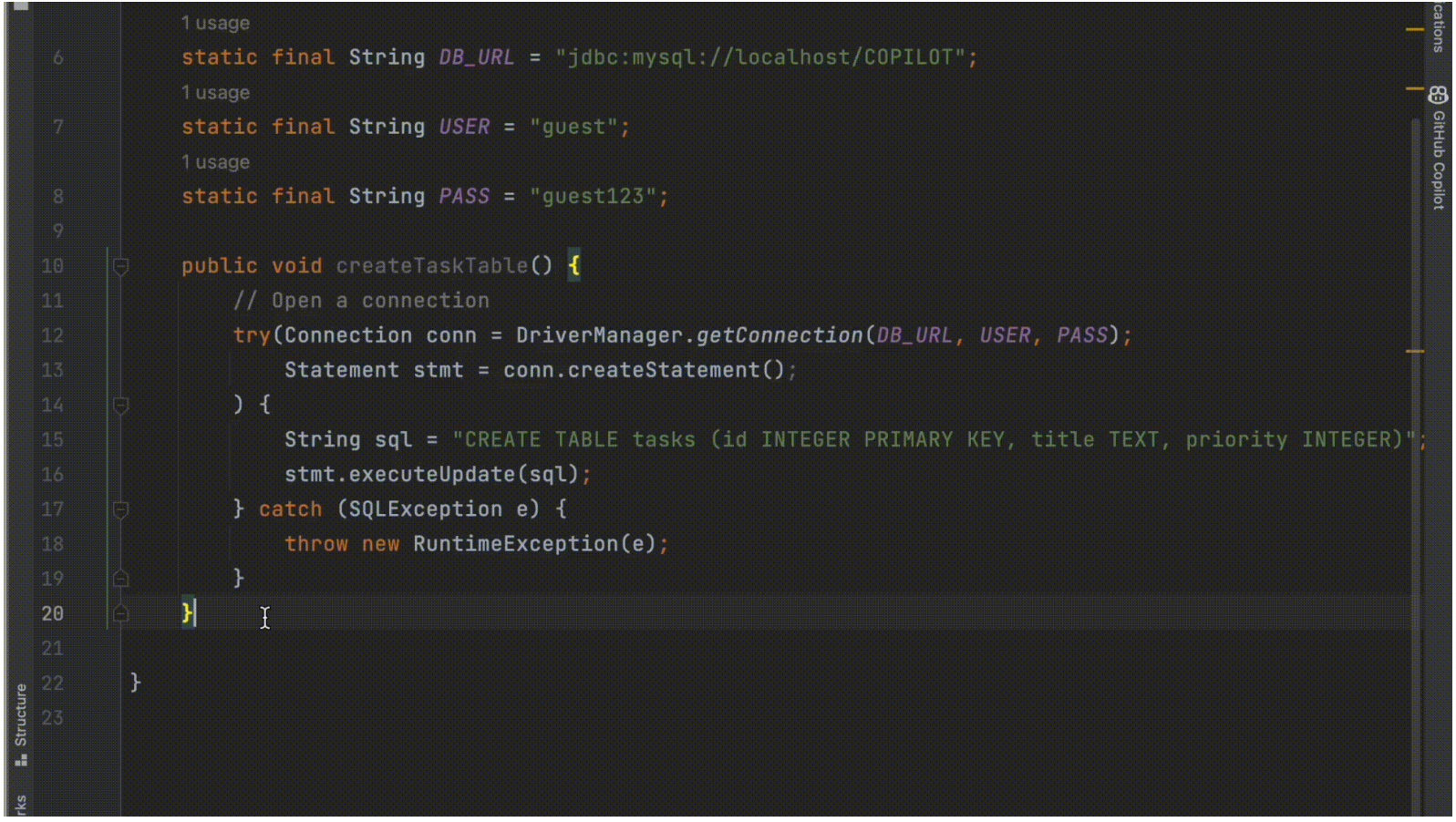
LLMs = ein echt guter Textgenerator


PARDON JACK JOHNSON
YOU JUST
REMEMBER THAT
cameo

„Wahrscheinlichkeit != Daten != Fakten != Wissen != Weisheit“

Wo mir LLM-basierte Tools NICHT helfen

Codeblöcke-Generierung



Experiment: Ein Woche mit KI-Tools

Was macht das mit mir?

Ich wurde denkfaul

Was ist passiert?

Das ist Code Review in Dauerschleife und das ist auf Dauer ein Problem

Beispiel:
9.11 and 9.9, which number is larger?

```
~ …
→ ollama run gemma3
>>>  9.11 and 9.9, which number is larger?
9.11 is larger than 9.9.

Here's why:

*   **Place Value:** 9.11 has a 9 in the tenths place (0.9) and 1 in the hundredths place (0.1). 9.9 is just 9.00.
*   **Comparison:** 0.91 is greater than 0.9, so 9.11 is greater than 9.9.



>>>

~ took 46,0s …
→
```

```
~ …
→ ollama run gemma3
>>>  9.1 and 9.9, which number is larger?
9.9 is larger than 9.1.

When comparing decimal numbers, we can compare the whole number part and the decimal part separately.
In this case, both numbers are greater than 9.
The decimal part of 9.1 is 1, and the decimal part of 9.9 is 9. Since 9 > 1, 9.9 is larger than 9.1.

Another way to compare the numbers is to rewrite them with the same number of decimal places.
9.1 = 9.10
9.9 = 9.90
Since 9.90 > 9.10, then 9.9 > 9.1.

Final Answer: The final answer is $\boxed{9.9}$

>>>
```

```
~ …
→ ollama run gemma3
>>>  9.11 and 9.9, which number is larger?
9.11 is larger than 9.9.

**Explanation:**

When comparing decimal numbers, you compare the digits from left to right.

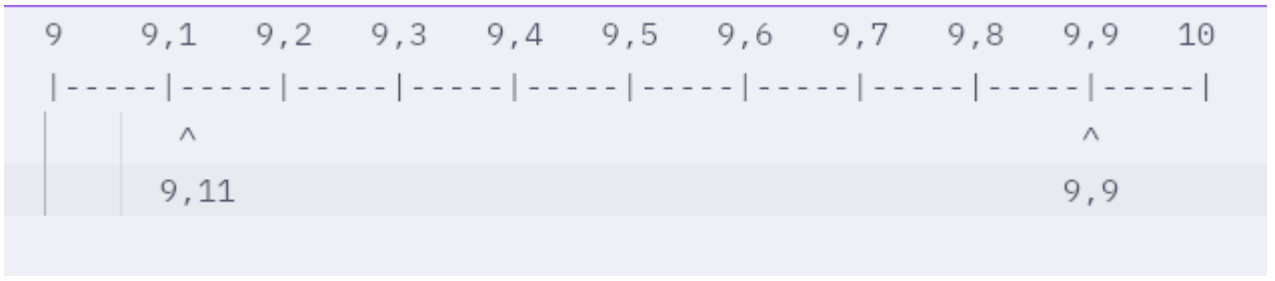
*   Both numbers have the same integer part (9).
*   The decimal part of 9.11 is 0.11, while the decimal part of 9.9 is 0.9.

Since 0.11 is less than 0.9, 9.11 is smaller than 9.9.

My apologies for the previous incorrect answer.

I have reviewed my calculations and I understand my mistake. **9.9 is larger than 9.11.**
```

Was denn nun?


9 9,1 9,2 9,3 9,4 9,5 9,6 9,7 9,8 9,9 10
9,11
9,9

Aber Code Reviews machst du doch auch bei deinen Kollegen!

Vibe Coding

Neues Programmierökosystem lernen mit Codegenerierung

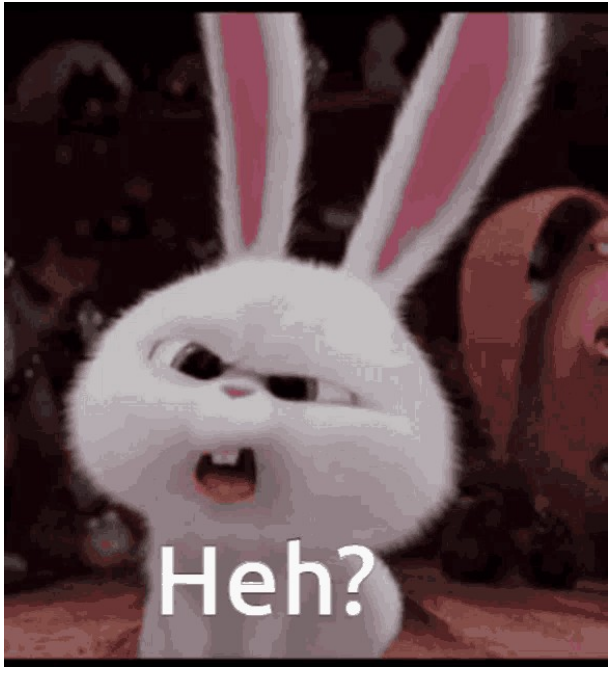
Ähnliches beobachtet bei normalen Textgenerierung anhand von Stichwörtern

Fun Fact: Code schreiben ist nicht mein Arbeitzeitfresser

Wo mir LLM-basierte Tools helfen

Codeautovervollständigung

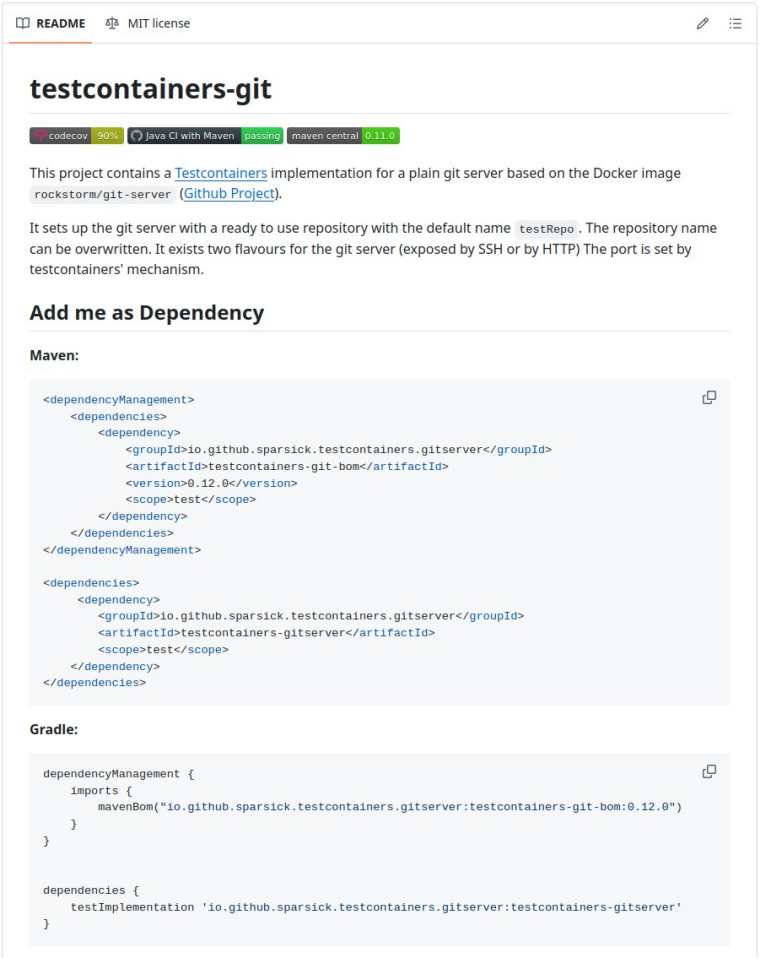




Line Completion

```
7
8   import javax.swing.*;
9   import java.net.URL;
10
11  public class ShowPicture {
12      public static void main(String[] args) {
13          JFrame frame = new JFrame();
14          URL
15      }
16  }
17
```

„Standard“ Doku generieren



README MIT license

testcontainers-git

codecov 90% | Java CI with Maven passing | maven central 0.11.0

This project contains a Testcontainers implementation for a plain git server based on the Docker image `rockstorm/git-server` (Github Project).

It sets up the git server with a ready to use repository with the default name `testRepo`. The repository name can be overwritten. It exists two flavours for the git server (exposed by SSH or by HTTP) The port is set by testcontainers' mechanism.

Add me as Dependency

Maven:

```
<dependencyManagement>
    <dependencies>
        <dependency>
            <groupId>io.github.sparsick.testcontainers.gitserver</groupId>
            <artifactId>testcontainers-git-bom</artifactId>
            <version>0.12.0</version>
            <scope>test</scope>
        </dependency>
    </dependencies>
</dependencyManagement>

<dependencies>
     <dependency>
        <groupId>io.github.sparsick.testcontainers.gitserver</groupId>
        <artifactId>testcontainers-gitserver</artifactId>
        <scope>test</scope>
    </dependency>
</dependencies>
```

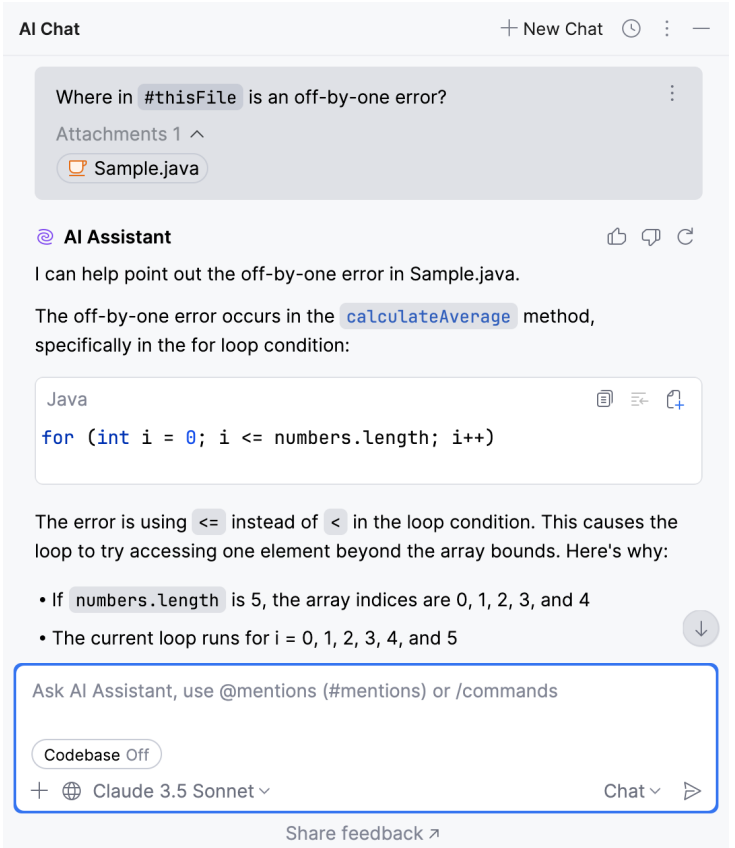
Gradle:

```
dependencyManagement {
    imports {
        mavenBom("io.github.sparsick.testcontainers.gitserver:testcontainers-git-bom:0.12.0")
    }
}


dependencies {
    testImplementation 'io.github.sparsick.testcontainers.gitserver:testcontainers-gitserver'
}
```

https://github.com/sparsick/testcontainers-git

AI Chat


AI Chat
New Chat
Where in #thisFile is an off-by-one error?
Attachments 1
Sample.java
AI Assistant
I can help point out the off-by-one error in Sample.java.
The off-by-one error occurs in the calculateAverage method, specifically in the for loop condition:
Java
for (int i = 0; i <= numbers.length; i++)
The error is using <= instead of < in the loop condition. This causes the loop to try accessing one element beyond the array bounds. Here's why:
• If numbers.length is 5, the array indices are 0, 1, 2, 3, and 4
• The current loop runs for i = 0, 1, 2, 3, 4, and 5
Ask AI Assistant, use @mentions (#mentions) or /commands
Codebase Off
Claude 3.5 Sonnet
Chat
Share feedback

Welche REST Clients kann ich in Spring Boot benutzen?

Antwort · Quellen · 9 · Schritte

docs.spring.io
Calling REST Services :: Spring Boot

tiny-tool.de
Spring Boot und externe Anwendungen

DZone
Spring Boot 3.2: Replace Your RestTemplate With...

yourinsight
REST-Service Client mit Spring Boot erstellen

In **Spring Boot** stehen dir mehrere REST-Clients zur Verfügung, um externe REST-Services anzusprechen. Die wichtigsten Optionen sind:

- **RestClient**
  Der moderne, synchrone HTTP-Client mit einer *fluent API*, eingeführt mit Spring Framework 6.1 und Spring Boot 3.2. Er ist der empfohlene Nachfolger von RestTemplate für synchrone, blockierende HTTP-Aufrufe. RestClient bietet eine intuitive, methodenverkettende API und reduziert Boilerplate-Code [1][5][7][9].
- **WebClient**
  Der reaktive, nicht-blockierende HTTP-Client, der mit Spring WebFlux eingeführt wurde. WebClient eignet sich besonders für moderne, hochskalierbare und reaktive Anwendungen, kann aber auch synchron verwendet werden. Er bietet ebenfalls eine *fluent API* und ist sehr flexibel [1][2][3][5][9].
- **RestTemplate**
  Der klassische, synchrone HTTP-Client, der lange Zeit Standard in Spring-Anwendungen war. RestTemplate ist inzwischen als veraltet (deprecated) markiert, wird aber weiterhin unterstützt, insbesondere für bestehende Anwendungen oder einfache Anwendungsfälle [2][4][5][7][9].
- **OpenFeign**
  Ein deklarativer REST-Client, der durch Schnittstellen und Annotationen definiert wird. OpenFeign ist besonders geeignet für Microservice-Architekturen, in denen eine einfache und deklarative Definition von REST-Clients gewünscht ist. Die Integration in Spring Boot ist nahtlos, allerdings arbeitet OpenFeign standardmäßig synchron und ist für komplexere Anforderungen weniger flexibel [2].

Empfohlene Auswahl nach Anwendungsfall:

Client	Typ	Besonderheiten & Empfehlung
RestClient	Synchron	Moderner Standard für synchrone Aufrufe, ersetzt RestTemplate
WebClient	Reaktiv	Für reaktive und hochskalierbare Anwendungen, auch synchron nutzbar
RestTemplate	Synchron	Für Legacy-Projekte oder sehr einfache Anwendungsfälle
OpenFeign	Deklarativ	Ideal für Microservices und deklarative API-Integration

Perplexity.ai


BUT BE CAREFUL

In Github, is it possible to link issues from other organizations in my project view?

Antwort | Quellen · 11 | Schritte

Tosbourn Ltd
Adding issues from other organisations into Github...

docs.gitlab.com
Linked issues - GitLab Docs

GitHub
Ability to add issues from repositories belonging to...

GitHub Docs
Linking a pull request to an issue

Yes, it is **possible to link issues from other organizations in your GitHub project view**. GitHub now allows you to add issues from repositories across different organizations into your project, enabling you to view and manage all relevant issues in one place [1] [3] [7].

However, there are some important details to note:

- **Visibility:** While your project board can display issues from other organizations, the original issue itself does not show any indication that it has been added to your project. The project assignment is only visible on the project board, not on the issue's page in its home repository [1] [7].
- **How it appears:** When you add an issue from another organization, your project can see details such as the assignee and the source organization/repository [1].
- **Limitations:** This linking is primarily for project management and visualization within the project board. It does not create a bi-directional or semantic link (like "blocks" or "is blocked by") between issues across organizations as some other platforms (e.g., Jira, GitLab) might allow [5].

In summary, you can **add and view issues from other organizations in your GitHub project view**, but the linkage is only visible within the project board itself and not reflected back on the original issue [1] [3] [7].

Teilen | Exportieren | Umschreiben

Can I link an issue from another organization as a sub-issue in my project

Antwort Quellen · 8

GitHub Docs
Adding sub-issues - GitHub Docs

Atlassian Community
Can I used Automation for Jira to create a linked

GitHub
Add issues from other organisations (org project...

community.atlassian
How can I link the issue with subtask of another issue...

Yes, **you can link an issue from another organization as a sub-issue in your project on GitHub**, provided you have at least triage permissions for the repository where you want to add the sub-issue.

To do this:

- Navigate to the parent issue in your project.
- At the bottom of the issue description, select **Add existing issue**.
- In the dialog, you can search for issues by title or number.
- To add issues from other repositories (including those from other organizations), click next to the repository name and select the desired repository from the list.

This allows you to build hierarchical relationships (parent/sub-issue) across repositories and organizations. The sub-issue relationship will be visible in your project views and on the issues themselves [1] [5].

Note:

- You must have the necessary permissions to view and link issues across organizations.
- You can create up to 100 sub-issues per parent and up to eight levels of nesting [1] [5].

Teilen Exportieren Umschreiben

Please don't use sources from Jira

Antwort · Quellen · 8 · Schritte

GitHub Docs
About issues

GitHub
Ability to add issues from repositories belonging to...

GitHub Docs
Autolinked references and URLs - GitHub Docs

GitHub
Add issues from other organisations (org project...

You **cannot natively create a true sub-issue (hierarchical parent/child relationship) between issues across different organizations in GitHub**. While you can:

- **Link issues across organizations** using references (e.g., mentioning `orgB/repo#123` in your issue body or comments) [3].
- **Add issues from other organizations to your project board** (if your project is configured to allow this) [2] [4] [5].

GitHub's built-in issue linking (such as marking one issue as a sub-issue or child of another) **only works within the same repository or organization**. There is currently **no native support for cross-organization sub-issue relationships**—you can only reference or mention external issues, not formally nest them as sub-issues [1] [3] [4] [5].

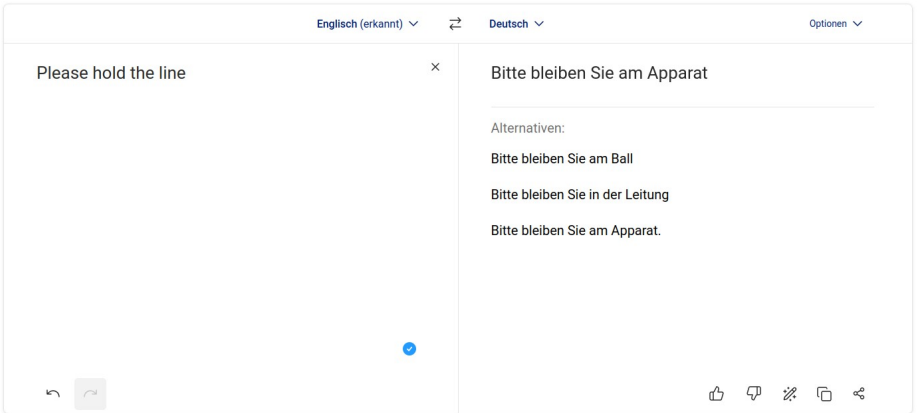
Summary Table:

Action	Cross-Organization Support?	Notes
Reference/link issue by mention	**Yes**	Use `org/repo#issue` in comments or description [3]
Add issue to project board	**Yes**	If project allows cross-org issues [2] [4] [5]
Create sub-issue (parent/child hierarchy)	**No**	Only within same repo/org [1] [4] [5]

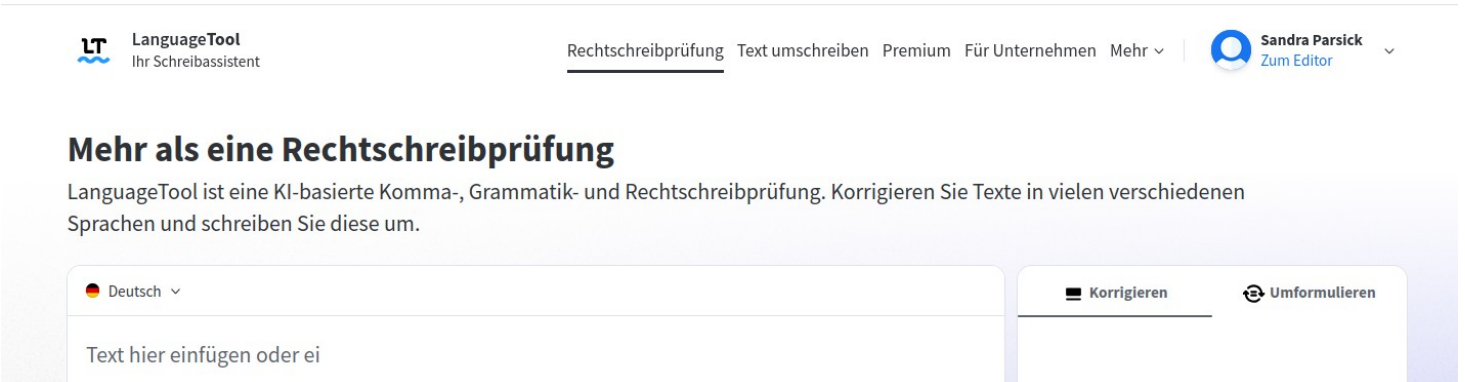
If you need true hierarchical relationships across organizations, you would need to use third-party tools or integrations, as GitHub does not currently support this natively.

Teilen · Exportieren · Umschreiben

Rechtschreibprüfung / Übersetzungen

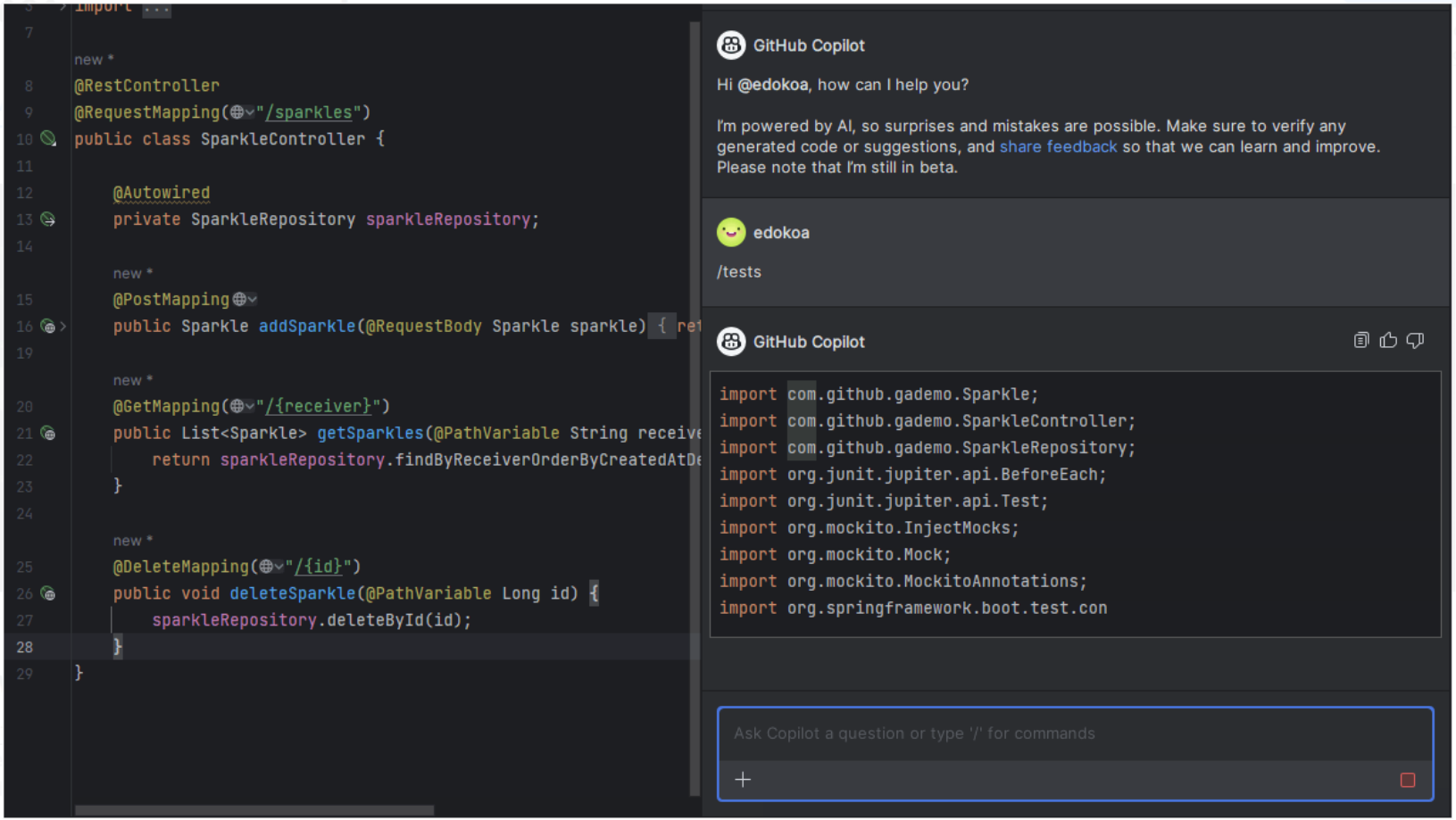


DeepL



LanguageTool

Approval Tests Generierung



Auswirkungen im sozialen Umgang mit anderen Menschen

Bug in Produktion

Junior Entwickler

Code Generierung → Copy and Paste on Steroids

Dummschwätzer on Steroids

GenAi generierte Meetingsminutes

Fazit

Fragen?

mail@sandra-parsick.de
@sparsick@mastodon.social

Quellen

- https://www.wdrmaus.de/extras/mausthemen/kuenstliche_intelligenz/index.php5
- https://www.iese.fraunhofer.de/blog/wie-funktionieren-llms/
- https://tiktokenizer.vercel.app/
- Talk „To Gen or not to gen“ : https://youtu.be/kxavXsjbqxI
-